

Niti Kanoongo

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EDUCATION

Vellore Institute of Technology, Bhopal

B.Tech CSE — AI & ML Specialization — CGPA: **8.66/10**

Aug 2023 – May 2027

Bhopal, India

EXPERIENCE

CovisionAI

AI/ML Intern

May 2026 – Aug 2026

Pune, India

- Built a benchmarking framework in *Python* to evaluate **3** frontier video generation models (Seedance, Kling, Runway) across quality, latency, temporal consistency, and prompt-adherence; findings directly informed production model selection.
- Engineered API integration layers for all 3 providers; designed prompt engineering workflows and automated test harnesses across **100+** generation configurations, cutting manual evaluation overhead by an estimated **40%**.
- Maintained version-controlled experiment logs and reproducible ML workflows; enforced data integrity and confidentiality standards across all benchmarking cycles.

Aspirant India Initiative

Website Developer & Reporting Manager

Mar 2025 – Jul 2025

Remote

- Architected and deployed a production web platform serving **40,000+** students across **500+** institutions; owned full lifecycle from system design through post-launch; led 3-developer team via sprint planning and code reviews.
- Shipped features covering **9** national SDG summit events across 5 months with zero production incidents; coordinated cross-functional requirements across PR, curriculum, and operations.

PROJECTS

SangamDrishti — Real-Time Emergency Command & Control System

[GitHub](#) — [Live](#)

React, Node.js, Express, Socket.io, RAG, Gemini 2.0 Flash, Vercel, Render

- Architected real-time crowd telemetry for Mahakumbh-scale events; engineered a persistent WebSocket pipeline (Socket.io) emitting live sector density and responder coordinates every **4s** across **10+** concurrent zones.
- Implemented geometric proximity dispatch algorithm computing live distance across all active responders, surfacing **top 3** nearest units for single-click deployment, eliminating human triage latency under crisis conditions.
- Built RAG-powered AI advisor (Gemini 2.0 Flash) grounded in **6** embedded SOPs with live telemetry injected per prompt; designed stateless backend with in-memory indexing; deployed to Vercel + Render.

Aurora — AI Productivity Dashboard

[GitHub](#)

Next.js 14, FastAPI, PyTorch, XGBoost, PostgreSQL, Docker, SHAP, Reinforcement Learning

- Designed modular full-stack system with async *FastAPI* backend and stateless workers for horizontal scalability; real-time inference endpoints with **60s** prediction caching reduce redundant compute across concurrent users.
- Built LSTM and XGBoost models for burnout detection and energy-level forecasting; integrated SHAP explainability; applied RL for adaptive task scheduling; used transformer embeddings for semantic task alignment.
- Optimized first-load JS bundle to **105 KB** via code-splitting and lazy loading; implemented JWT authentication with modular API design and version-controlled training workflows for reproducible model iteration.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, Bash

Systems & Tools: Git, Linux, Docker, GCP, AWS, Azure, PostgreSQL, MongoDB, Firebase, Redis

Frameworks: FastAPI, Node.js, Express.js, Socket.io, Next.js, React

AI/ML: PyTorch, TensorFlow, Scikit-learn, XGBoost, SHAP, LSTM, CNN, RAG, Gemini API, Prompt Engineering, OpenCV

Concepts: Distributed Systems, Real-Time Streaming, Async Concurrency, REST APIs, WebSockets, System Design

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (May 2026) | Oracle Cloud Infrastructure 2025 AI Foundations | Applied ML in Python – University of Michigan | Full-Stack Web Dev – Udemy

ACHIEVEMENTS & LEADERSHIP

- **Competitive Programming:** LeetCode **1415** | CodeChef **1060** | **900+** problems solved | TCS CodeVita 2026 Global Rank **8700**
- **Programs:** Amazon ML Summer School 2026 (Selected) — 6-week intensive: Supervised Learning, Deep Learning, Probabilistic Graphical Models, Bandits & RL, Causal Inference, NLP
- **Hackathons:** Finalist — i4c Dynamic Hackathon (Top **400** nationally) | Shell.ai | Redrob AI 2026 | Cyber Police
- **Open Source:** GirlScript SoC 2024 & Hacktoberfest 2025 Contributor | FOSSEE Fellowship — IIT Bombay (Scilab IPT)
- **Leadership:** Sec. General, Cisco Community VITB | Microsoft Learn Student Ambassador | Chairperson & EB at **30+** MUNs | Freelance Web Developer (2 paid client deployments)
- **Honors:** Dell Champs AIR 2 (2016), AIR 3 (2015) | SilverZone French Olympiad AIR 23